Exercice 2:

a) Model A:

$$\phi_A(x) = 1 - 0,9\,x - 0,09\,x^2 = 0$$

$x_1 \simeq -11$ and $x_2 \simeq 1,01$

$|x_i|_{i=1,2} > 1$ so $\{y_t\}$ is stationary and it's an

ARIMA$(2,0,0)$ = AR(2)

Model B: $y_t = y_{t-1} + e_t - 0,1\,e_{t-1}$

$\phi_B(x) = 1 - x = 0 \iff x = 1$ and $|x| = 1$

so $\{y_t\}$ is not stationary

$$\nabla y_t = e_t - 0,1\,e_{t-1}$$

(*) $E[y_t] = \underbrace{E[e_t]}_{0} - 0,1\,\underbrace{E[e_{t-1}]}_{0}$

$= 0$

Using exercice 1 we have:

(**) $\text{Cov}(y_t, y_{t-h}) = \begin{cases} \sigma_e^2 + (0,1)^2\sigma_e^2, & \text{if } h = 0 \\ -0,1\,\sigma_e^2, & \text{if } h = 1 \\ 0, & \text{o.w} \end{cases}$

hence $\{\nabla Y_t\}$ is stationary.

conclusion: $\{Y_t\}$ is an $ARIMA(0,1,1) = IMA(1,1)$

b)

Model A :

GLP representation of $\{Y_t\}$ (already stationary) is :

$$Y_t = \sum_{j=0}^{\infty} \psi_j \, e_{t-j}$$

with $\psi_0 = 1$; $\psi_1 = 0,9$;

and $\forall\, j \geq 1$, $\psi_j = 0,9\,\psi_{j-1} + 0,09\,\psi_{j-2}$

Invertible representation of $\{Y_t\}$

$$e_t = Y_t - 0,9\,Y_{t-1} - 0,09\,Y_{t-2}$$

Model B : $\{Y_t\}$ is not stationary so we use $\{\nabla Y_t\}$

GLP representation of $\{W_t\} = \{\nabla Y_t\}$ is :

$$W_t = e_t - 0,1\,e_{t-1}$$

Invertible representation of $\{Y_t\}$

we have $W_t = (1 - 0,1B)\, e_t$

$$\Rightarrow \quad e_t = \frac{1}{1 - 0,1B} W_t$$

but $$\frac{1}{1 - 0,1x} = \sum_{j=0}^{\infty} (0,1x)^j$$

so replacing the B operator yields

$$e_t = \sum_{j=0}^{\infty} (0,1)^j W_{t-j}$$

hence $\pi_j = (0,1)^j$